Tangents to a circle

1

a $OB < OC$

b Point B

c The angle is a right angle.

d Right angle

2

a Note that a tangent to a circle is perpendicular to the radius drawn to the point of contact. This means that $\angle OBA = 90^\circ$ and $\angle PBA = 90^\circ$. Observe tha $\angle OBP = \angle OBA + \angle PBA$. By substituting the values of two angles we get, $\angle OBP = 90^\circ + 90^\circ = 180^\circ$. Hence, $\angle OBP$ is a straight angle.

b They are collinear.

3

a $x = 90^\circ$. A tangent to a circle is perpendicular to the radius drawn to the point of contact.

b $x = 125^\circ$. Since a tangent to a circle is perpendicular to the radius drawn to the point of contact, $\angle OAB = 90^\circ$ and $\angle OCB = 90^\circ$. Note that the interior angle sum of a quadrilateral is 360°. So, $x + 55^\circ + 90^\circ + 90^\circ = 360^\circ$ which gives $x = 125^\circ$.

c $a = 31$. Since a tangent to a circle is perpendicular to the radius drawn to the point of contact, $\angle OPS = 90^\circ$. Note that the angles forming a right angle are complementary. So, $a^\circ + 59^\circ = \angle OPS$ which gives $a = 31$.

4

a $a = 59$ **b** $b = 18$ **c** $y = 93$ **d** $y = 123$

5

a 98°

b 262°

6

a $(180 - x)^\circ$

b $(x + 180)^\circ$

Additional questions

7

a No

b Yes

8 Yes

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9 $x = 27^\circ$

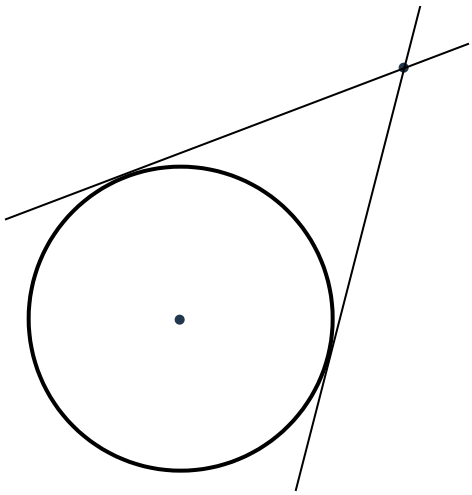
10

a $y = 28$

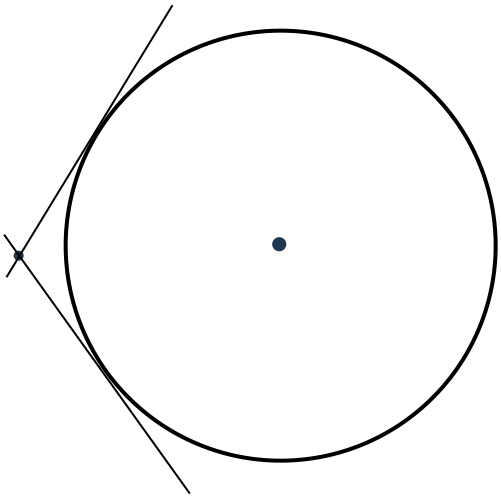
b $x = 74$

11

a



b



12

a $x = 58$

b $x = 24$

13 $OK = 21$

14 34 cm

15 78

16 $KL = 68$

17 $FH = 8$

18 $BD = 26$

19

a $x = 19$

b $XT = 15$

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20 **To prove:** $\overline{AC} \cong \overline{BC}$

	Statements	Reasons
1.	$\overline{AC}$ and $\overline{CB}$ are tangent to circle O	Given
2.	$\angle CAO$, $\angle CBO$ are right angles	Definition of right angles
3.	$\overline{OA} \cong \overline{OB}$	Definition of radii
4.	$\overline{OC} \cong \overline{OC}$	Reflexive property
5.	$\triangle CAO \cong \triangle CBO$	HL congruence postulate
6.	$\overline{AC} \cong \overline{BC}$	CPCTC

21 The radius of the cone must be equal to the radius of the scoop, and the sides of the cone must be tangent to the scoop at its diameter.

22 We know that $\overline{RM} \cong \overline{RL}$ because they are both tangents to the circle G. Similarly, $\overline{RL} \cong \overline{RN}$ since they are both tangents to circle P. Using the transitive property we get that $\overline{RM} \cong \overline{RN}$.

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